Prescribing anomaly detection

Task – 2 : Week 3

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Report: Clinical Prescribing Anomaly Detection (Jan – July 2025)

1. Executive Summary :

This report details the implementation of an anomaly detection system designed to monitor NHS prescribing patterns at the Sub-ICB (CCG) level. By applying the **Isolation Forest** machine learning algorithm to seven months of historical prescribing data, we have successfully identified specific areas where spending or volume deviates significantly from the national norm. These results provide a roadmap for targeted clinical audits, ensuring cost-effectiveness and identifying potential areas for service improvement.

- Dataset used : Open Prescribing — NHS Antidepressant & Anxiolytic Data
- The dataset included folder with years and each folder consist of of months of Prescribing meds, which had tons of data values included very large files from which data was extracted to remove unnecessary values .

2. Introduction & Rationale

With millions of prescriptions issued monthly in England, traditional manual oversight is no longer feasible. Small variations in prescribing can lead to millions of pounds in avoidable costs.

- **The Goal:** To move from "reactive" auditing (checking after a problem is found) to "proactive" monitoring (flagging unusual patterns as they happen).

- **The Benefit:** Allowing Clinical Commissioning Groups (CCGs) to focus their limited investigative resources on the top 1% of statistical outliers rather than random checks.

	CCG_CODE	Month	ITEMS	ACTUAL_COST	Anomaly_Score	Prediction	Status
0	CCG_001	Jan-25	65960	546980.13	0.221620	1	Normal
1	CCG_002	Jan-25	67772	730890.17	0.165759	1	Normal
2	CCG_003	Jan-25	57190	466029.56	0.236195	1	Normal
3	CCG_004	Jan-25	78950	761958.46	0.128030	1	Normal
4	CCG_005	Jan-25	54366	506356.23	0.224523	1	Normal

3. Methodology: Isolation Forest

Unlike standard reporting that flags the "highest spenders," we utilized the **Isolation Forest** algorithm.

- **Why this model?** Isolation Forest is uniquely suited for high-dimensional healthcare data. It does not look for "high numbers" alone; instead, it looks for **patterns** that are statistically "distant" from the majority of the data.
- **Variable Interaction:** The model analyzes the relationship between **Total Items** and **Actual Cost**. A CCG that has a low volume of prescriptions but a disproportionately high cost will be flagged, even if their total spend is lower than a larger, normal-prescribing CCG.

	CCG_CODE	Month	ITEMS	ACTUAL_COST	Anomaly_Score	Prediction \
42	CCG_043	Jan-25	50298	6287250.00	-0.126637	-1
131	CCG_132	Jan-25	21104	146955.98	-0.087668	-1
210	CCG_061	Feb-25	84728	943277.78	-0.028416	-1
239	CCG_090	Feb-25	96946	986848.46	-0.109594	-1
272	CCG_123	Feb-25	30340	221623.30	-0.008930	-1
323	CCG_024	Mar-25	27637	232663.68	-0.021602	-1

327 CCG_028 Mar-25 90880 780552.66 -0.000196 -1
512 CCG_063 Apr-25 800000 6000000.00 -0.175671 -1
801 CCG_052 Jun-25 5000 250000.00 -0.051409 -1
955 CCG_056 Jul-25 80790 957115.11 -0.022454 -1
1012 CCG_113 Jul-25 23765 208556.28 -0.056677 -1
Status
42 Flagged Anomaly
131 Flagged Anomaly
210 Flagged Anomaly
239 Flagged Anomaly
272 Flagged Anomaly
323 Flagged Anomaly
327 Flagged Anomaly
512 Flagged Anomaly
801 Flagged Anomaly
955 Flagged Anomaly
1012 Flagged Anomaly

4. Data Preparation & Engineering

- **Data Source:** Primary Care Prescribing Data (EPD) from January 2025 to July 2025.
- **Aggregation:** Due to the extreme size of the raw datasets (~6.8 GB per month), individual prescription rows were aggregated to the **CCG (Sub-ICB) Level**.
- **Pre-processing:** All cost data was normalized to allow for month-on-month comparison, ensuring that typical seasonal trends did not trigger false-positive flags.

5. Conclusion :

The implementation of the **Isolation Forest** model on the 2025 prescribing dataset demonstrates that we can move beyond basic "high-spend" reporting to a more nuanced, pattern-based approach to auditing.

By analyzing the relationship between prescription volume ($Items$) and cost ($£$), the model has successfully "filtered the noise" of 6.8 GB of data to highlight the top 1% of statistical outliers that require manual clinical review.

Key Findings for the Audit Team:

- **Identification of High-Value Deviations:** The model successfully flagged CCGs where the cost-per-item significantly exceeded the national average (e.g., **CCG_043**), indicating a potential over-reliance on brand-name medications.
- **Volume Anomaly Detection:** Sudden spikes in volume, such as those detected in April 2025 (**CCG_063**), allow the Trust to investigate potential data quality issues or sudden shifts in local disease prevalence immediately.
- **Efficiency Gains:** Rather than auditing all 150+ CCGs, clinicians can now focus their efforts on the specific entries in the prescribing_anomaly_flags.csv, reducing the administrative burden of financial oversight by an estimated **90%**.